Quick access toolbar. (Autosave, Mail box)
Ribbons Home, Ingest.
Features. Cond^n form., Bold.
Categories Number, Font, Style.
Name box A1
Formula box dual broadcasts of Excel.
Cells. ▭ → cell selector.
Range. ▭ → fill handler.
Sheet name.

features : less time consume, fail at complex sc., not al
function : more time , complex scenario ✓, self up
data

① Flash fill : Merging or splitting of data.
(CTRL+E)

name	class.	Merge-column.
Aman	2	Aman2
Ajay	4	
Diya	3	

▭ ✗ select → home.
→ Editing → ▭ Flash fill.
or.
▭ ✗ CTRL+E ✓.

Date Error : 19/09/18 | Date: 19 | Month: 09 error =9 error. | Year.
3/11/17
4/01/18.
↓ ↓
'09
'11
'02 ← CTRL+E

Flash fill fail : • 19/03/18
14/11/22
March.
CTRL+E Co
• Currency
• Calculation not possible. (use Func & formula)

② Remove Duplicates :

Select full data. → Data (Ribbon) → Remove Dup.

Customer ID
Eg: Row having same customer id.

Single column
Multi column → Customer ID | Gender
Remove repeating customer id only if gender also same.

③ Text to Column.

- Fix width.
- Delimited.

DD
19/09/18
13/03/17

Date Month Year.

Select → Data (Ribbon) → text to column → Fix width
(full data/column)

Select destination

Select Range.

19	09	18
13	03	17

Finish

date	Month	year.
19	09	2018
13	03	2017.

- Text to column • Delimited.

→ In fix width Range used ⇒ 19 17 | 09 03 | 18 17

→ In delimited sign = delimiter

Eg: Pawan Sharma. / Khushi Chaudhary → Select → text to column → Delimiter = Space. → Select destination

Eg: for date. delimiter = /

Fail of Text to column.

• Extract of value : apple cost 1.2 USD
42 INR is auto fail.
USD 95 is his loss

• Date
3/4/15
4/2/18

Month Date Year.
18
17.

Apply Delimiter → Error (Have to replace year or column).

Sheet Management - ① Right click → Rename.
→ Add sheet
→ hide / unhide
→ move / copy (Duplicate)
→ Delete.

Sheet 1 Sheet 2 ⊕

② Auto fit column width.
Aman Singh / Khushi → select column or full data. (CTRL+A)
↓
home → format → auto fit column width.
auto fit Row width.

③ Selecting full Rows:
select → CTRL + Shift + →
↓
i. apply desired format like alignment ≡ B, I
ii. to format all rows → format painter (clipboard)
continue (i) → CTRL + Shift + ↓ (OK).

④ Copy Paste.
Select full data or table by short cuts.

- Copy + Paste → full formats + formula.
- Copy & paste value → only value paste
- copy as image → for print also.
- Copy + Linked paste → D main table = D Paste.
- Copy + paste special (Skip blanks)

New copy paste skip old.

New: 3 → old: 3333 → update (only some cell)

⑤ Copy and paste by performing operation on them.
old: 750, 420
New: 750, 420
last new must be 835, 495.

paste old, old, new
- select cell of 114% → copy.
- select the column where to paste ⑤ paste special.
  → Multiplies.
  → Values.

⑤ Freeze unfreeze
Select any data cell → View → ❄ freeze top Row
❄ freeze top column
❄ freeze Panes:

all above row
all left column
Selected cell

⑥ Merging.
Select → home → merge & center.

Select → home → merge & center
↓
alignment → ≡

OR → OK → Select → format painter
all column
(CTRL + Shift + ↓)

Filter

- select all data → home → editing → F&S → Filter.

Name ▼ Gender ▼ Class ▼ → such sign appear.

* Misaligned

Name Gender Class

→ select all data → Apply filter at Right Place
→ select any cell of data → Apply filter, at Right Place.
→ select empty area along with data → Apply filter, not Right Place.

Date filtering (before 7 Jan)

Apply filter → Contact_date ▼ ① → Date filter → before
1-Jan-23
2-Jan-23
② Manually choose.
Enter 01/07/2023

Time filtering (b/w 10:00:00 AM & 12:00:00 PM)

Select all data → filter
time_column ▼ → Number filter → between → ≤ 10:00:00 AM
≥ 12:00:00 PM

Sorting

Way ① Column Data
1-Jan-23
2-Jan-23
→ select → home → F&S → Sort.

Way ② select all data → filter → ▼ ▼ ▼
Sort available

Custom Sorting

Multiple sorting: transaction based on agent & Morning to Evening

Select all data → home → F&S → Custom sorting
Add value ④
sort by Agent-name order A-Z
sort by Contact-time order Small to Large

Marks	Name	Subject
4	Aman	Physics
7	Ankita	ML
8	Kavita	ML
9		
2		

Marks	Subject
>5	ML

← Criteria.

Advanced filtering ①

Filter on basis of more data.

- Select main data → Data → Adv Filter
- Filter list in Place.
  - distance Range: Main data (select)
  - Criteria Range: select Criteria data OK.

Advanced Filtering ②

Sheet 1 — Main data
Sheet 2 — Criteria data — A1

- Select A1 (Sheet 2)
- adv. filtering
- Copy to another location
- List Range: Select A1 then open sheet 1 → Select all sheet (CTRL A)
- Criteria Range → Criteria Table (Sheet 2)
- Copy to: A1 (Sheet 2) → data will come in A1 of sheet 2.

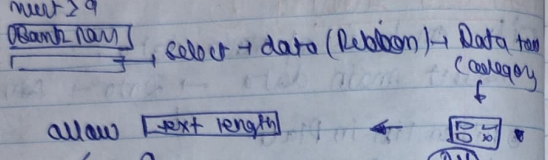
Sheet 1 Main data | Sheet 2 Criteria data | Sheet 3 A1

- select A1 of sheet 3
- adv filtering
- Copy to another
- List Range: A1 (Sheet 1) → Main data select all.
- Criteria Range: Criteria table (Sheet 2)
- Copy to A1 of sheet 3 →

Data Validation :

Can enter data in cell of any particular range if it is valid

① Length.



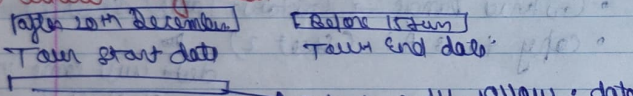
make such validation on full column.
→ make data validation on particular cell.
→ copy (CTRL+C) that cell.
→ Select all desired cells / columns.
→ CTRL + V.

OR.

apply DV on cell → format painter → select rest cell/col

---

② allow only date.



Setting
allow
:

title
Input message :
title :

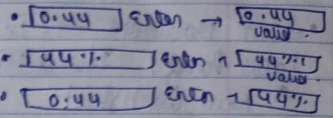
Error Message
Title :
Error :
Msg

③ allow only time

data validation → allow : time
data : >
Start time : 11:00:00 PM.

④ allow on '/'.

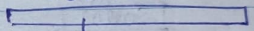
DV → allow : decimal.
: greater than
: 0.35



⑤ allow only list :

Pizza, burger, fries only
order item

create from table H
order item.



DV → allow : list
Source : pizza, burgers, fries

DV → allow : list
Source : select full H table / column.

→ order item 1 order item 2

⑥ Data allow Custom (AI chatgpt).

STU XXXX
STUDENT ID

Current Date
Date

@gmail.com.

F7 :

DV → allow : custom.
formula : chatgpt.

Prompt : I need validation formula for excel. for student ID.
Eg : STU4926 → first 3 STU next numeric.
This is for cell adress F7 & will copy for other rows as well → Give me custom code for this situation

Pivot Table, Data Analyses

Source data → backend (calculation / conversion) → front end (display).

- Select Main data → Insert → Pivot Table.

Table Range: [Select Main sheet]
- ◉ New worksheet] your choice.
- ○ Existing

Location: Sheet 3 or sheet name.

Pivot Table field

- Select column : Sales, Price, x yz.

Drag: Filter, Row, Column, Value.

Drag → Sales → Sum: Sales 502697

→ Rename: Sum of Sales
→ Sum, Product, Avg, Max, Min, Count

- Select column : account_name, Sales, price, x yz.

Drag: Filter, Row, Column, Value.

- Sales → avg
- price → sum

Account_name	Avg of Sales	Total
Pawan	2002	926
Khushi	2001	
Total		

- Filling column.

only Row & value ⇒

Account	Sales
Pawan	
Khushi	
Total	

Row & value & column / Cover type ⇒

Account	Cover type: ML Sale	Cover type: AI Course
Pawan		
Khushi		
Total		

- Copy column, cell → paste in pivot → not allow
make change direct in pivot like rename etc.

- Sorting in pivot table.

account, Sales

Way 1 → Right click → sort

Way 2 → Click → More sort option
- ○ Ascending A to Z by Product name.
- ○ Descending Z to A by Product name.

Sales ok

- Label filter
* Contains.

Eg: filter out Sharma from list.

account, Sales → Click → Label filter → Contain → Sharma

- Value filtering
*

Eg: filter out Sales > 1000

account, Sales → click → Value filter → > → Sales > 1000 OR

- Report filtering

back to filling column. → Filter → Cover type.

Table:

Cover type	Account	Sales
	Pawan	
	Khushi	

Take out only for ML Sales

When Row = date → Row = Day( ) / Month( ) → Rename: Day name

Nov / December → Nov ; December

1- Nov - 2021
2- Nov - 1

Number formatting

① Product | Sales → select → Right → No. formatting → Number : Decimals

19.ww
32.ww

2.324698 → 2.32
19.624692 → 19.62
32.458674 → 32.45

② 0.476 ... → No formatting → Percentage → Decimal : 2

Analysis ①

Row = date
Column = Overall Sale, Avg Rating
Notice ⇒ ⓐ when sale was more rating was ~~bad~~ more
when sale was less rating was less.

Analysis ②

Top ③ Sales & top ① Rating's.

Day | Sales | Avg Rating → select → Value filter → Top 10

Top | 3 | Sales

OK.

copy pivot table → Day → select → Value filter → Top 10

Bottom | 3 | Rating

→ Day | Sales | Avg Rating

Day | Sales | Avg Rat

PTA → Pivot Table Analyze (Ribbon)

Δ in Main data.

Δ in Main Data → not change in Pivot.

→ Make Δ in Main Data → open Pivot table → select any cell.

→ Pivot Analyze (Ribbon) → Refresh / Refresh All.

Δ in Main data (Addn of new row / column)

Select Pivot table any cell → Pivot Table Analyze (Ribbon)

→ Change Data Source.

→ ⊙ select a table or Range | Select full table | OR select

OK

→ Refresh.

Analysis ③ (Grouping)

Row = Rating
Column = Sales.

→ Rating | Sales

1
2
3
4
5
6
7
8
9
10

Select 1-2-3 → PTA & → Group Selection
Select 4-5-6 → PTA → Group Selection
Select 7-8-9-10 → PTA → Group Selection

→ Rating | Sales

⊕ Group ①
⊕ Group ②
⊖ Group ③

7
8
9
10

→ Grp in Row = Rating | Rating 2 automatically → Overall Rating & Rename Grp ① Grp ②

→ Rating 2 | Sales

Table 1 → avg sale (account wise)
Table 2 → model vs Avg price.
Table 3 → type vs sale amount

but want only detail of these 3 tables of specific category or anything.

Slicer (Back to report filter once)

→ this is for one table only
→ for all table in sheet, it fails
→ Slicer.

Select → PTA → Insert Slicer.

Select for what you want to do. Eg: Country

OK.

This is only for selected Pivot tables.
→ to make it for all

Select Slicer box on sheet → Slicer (Ribbon) → Report Connection

already made

Select all tables

Select → Insert Timeline → Contract_date

This will look like

Jan Feb March ... Dec month

Click on specific Month for specific month data in these 3 tables.

Conditional Formatting. Home → Conditional Formatting

① Data bars.

• Select the value column → home → CF → Data bars

• " " " → Data bars → More Rules → for color, format, & Show bars only

$ Show bars only

$ Customize data bar (Data bar only for total sale ≥ 25000

• Select total sale column → home → CF → Data Bar → More Rules.

Apply:
⇒ ⊙ on cells showing "Total sales" value for "Product name"

Minimum : 25000
Value : Number → OK ⇒ < 25000

② Color Scales.

Sales → Least sale, Most sale.

Select column data → home → CF → Color scale → Select any type or More Rules

$ Customize Color scales.

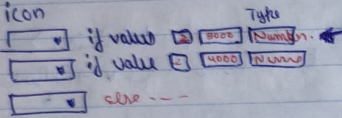
Example: sales data → 3 scale (Color scale) selected → More Rules.

data
20
65
50
74 → Must be white not 50.
90

	Mini	Midpoint	Maximum
Type	Lowest v	Number	highest value
Value		50	
Color	Red	white	green

③ Icon sets

Select column data → Home → CF → Icon sets → Select as per req / More Rules



④ High light cell Rule

Example: out of Customer ID : highlight Duplicate
: =
: text contain
: ≤ ≥ < > =

• select data column → Home → CF → highlight cell Rules

Duplicate ▾ value with Red ▾ OK

⑤ Top / Bottom Rules

to highlight → top 10 or xy item
bottom 10 or xy item
top 10% or xy% item
bottom 10% or xy% item
Above Avg
Below Avg
More Rules

Select column data → Home → CF → Top/Bottom

Can be applied two or more at one time
→ Top 3 → Red
Bottom 3 → Green

Print & layout

Printing in pdf.

① → Print → Setting
Page Orientation → landscape
Type → A4
Margin → Narrow
Scaling → Fit sheet on one Page

② Print Area

only this part to print

• select Print Area → Page layout (Ribbon) → Print Area ✓
→ Go to Print → apply setting as per req. ☺
print selection

③ Just remember keys

Merge and centre (combine merging)

Print area → apply page break → print area

Go to Print Page ① Page ② } separate page.

④ Print titles

→ Print → setting (all orientation check)
→ Print titles (in Page layout) → Row to repeat at top (select R①)

Page 1 Page 2 Page 3

to apply gridline in Printing # → print title → gridlines ✓

⑤ To print full (Contain applied Tables)

Headings

not to include.

• apply Print area (don't also include heading)
• apply Page title : heading = row at top
• vertical centre all print
horizontal centre

Formula & function

- Formula: =A1+A2
- Function: =sum(A1,A2)

Sum → data → cell (eg: C4)
will have value of applied func from data (selected Range here)

=sum(A1, B2, C1) → here A1, B2, C1
- can be selected cells
- can be Ranges selected
- can be selected cells of diff sheet
- can be selected ranges of diff sh

Just = sum(select cell/Range , (also more Range - -).
(apply comma)

▭ → select cell → Formula (Ribbon) → Insert Fun
→ select Sum → input N1 (cell No.) (select)
N2 (cell No.) (select)
⊕ OK.

=A1−C2+G4 (cell) OK.
select C2 → =A1−C2+G4
F4 press → value of C2 appear.

Avg
Range / cell of same or diff sheet.
= Average (, ,)

Count / Count A

X	Y	Z
—	2	Amazon
—	4	Microsoft
—	9	Apple
—	6	Google
—	8	Meta

= Count (B2:B6)
= CountA (C2:C6)
→ Count (, ,)
CountA (, ,)

they ignore empty cell → =count(B2:B6) → 4 only.
CountA dont ignore column header (Eg: account-name) if selected within range. So, select data wisely.

Max
data → max → value out of selected values.
Range / cell of same or diff sheet.
= Max (, ,)
= Min (, ,)

Absolute or Fixed Referencing

	X	No of order	Price
①	Pizza	2	100
	burger	3	50
	chowmin	4	25

Total → = B2 $ C2 ok.
- Now copy the cell (D2)
- Paste it in all next cells below D2.

Total → Paste → 200 → Paste → 300 / 150 / 100

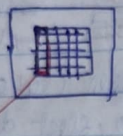
② Absolute & fixed Refer...

Total: = B2 $ C2 copying this makes value = B3 $ C3, = B4 $ C4 in pasting.

	A	B	C	D
	X	Order	Price	Total
2		2	100	
3	B	3	50	
4	C	4	25	

if entered =B3 $ (C3 − (B3 $ C3) $ B1
while pasting it B1 → B2 that is wrong. to fix B1.
= B3 $ C3 − (B3 $ C3) $ $B$1
B1 fixed

Named Ranges



• Select full column (header included)

↓

Name box : Enter ⇒ xyz-column Enter

Know unselect that → Name box ▼ → xyz-column → Select that part now.

=COUNTA(xyz-column)

another way to make such named ranges

Select Range → Formulas → Named Manager → New
→ Define name → Name: ... → OK

to delete it → Named manager → Select Name → Delete

IF

=IF(Validate logic, what if logic runs, what if logic fails)

Eg:

food quality	company	will attend?
Good		
Avg		
Bad		
Good		

=if(C2="bad","not go","go") Enter

copy the cell → Paste in Rest → OK.

And / or:

Price A	Price B	Price C	xyz

=And(B2>C2, C2>B2) → if true then TRUE else false.

SumIf

=SUMIF(Criteria Range, "Criteria", sum Range)

Example

Fruit	Price	Month
Apple		Jan
Banana		Feb
Apple		M
Pineapple		Dec

Select cell → (Random)

=SumIf(A2:..., "Apple", B2:...) OK

SUMIF + wild card

=SumIf(A2:..., "___", B2:...)

"apple" → fruit must be apple

"*apple*" → must have apple in text (pineapple apple)

"*apple" → must have apple at end: pineapple or an apple

"?????" → month starts with J

"J*")

CountIf

=SumIf(A2:..., "apple")

↓

3 apple purchase were made (3 apple in fruit column)

IFS

IFS = IFS(Logic, Value if true, Logic, Value if true..)
output

=IFS(B2>6, "Greater than 6", B2<3, "it is less than 3")
↳ check B1>6? if not, check B2<3.

Sumifs =Sumifs("Select Range", Value, "filter1", select full column, "filtered1 object", filter2, column, filtered2 object)

Eg: =Sumifs("B1:B7", price, C1:C7, "Amazon", D1:D7, "March") (# "March" = J7)

- This will give sum of price of all amazon of march.

Eg: =Sumifs(B1:B7, E1:E7, "Kiwi", F1:F2, "≥ 2022-07-10", F1:F2, "≤ 2022-07-25")

Sum of price of Kiwi sold b/w 2022-07-10 & 2022-07-25.

Average IFS

=Averageifs("Select Range", Value, filter1, Column range, "filtered1 object", filter2, "filtered2 object").

Eg: averageifs(D1:B7, price, K1:K7, vendor name range, "*O*", vendor name have O letter)

=MaxIFS(..., F1, F01, F2, F02)
=MINIFS(— — — —)

=Left =Right =Mid

Aman Sharma

Extract first 3 letter : =left(A2, 3) = Ama.
Last 3 letter : =right(A2, 3) = Hma.
Start at 2 and continue 6 : =mid(A2, 2, 6) = man Sh

=Upper(..) =Lower =Proper.

Aman Sharma.
= upper(A2) = AMAN SHARMA.
= lower(A2) = aman sharma.
= proper(A2) = Aman Sharma.

Concat and TextJoin

Aman | Sharma | Jane Street

= Concat(A2, B2, C2) ⇒ AmanSharmaJane Street

= Concat(A2, "=", B2, "=", C2) ⇒ Aman = Sharma = Jane Street

= TextJoin("=", TRUE, A2, B2, C2) ⇒ Aman = Sharma = Jane Street

Aman | | Jane Street (Empty cell)

= TextJoin("=", TRUE, A2, B2, C2) ⇒ Aman = Jane Street

= Concat(A2, "=", B2, "=", C2) ⇒ Aman = = Jane Street

= today()

today date | =today() | 31/12/2024

Day Month Year hour minute second (Extract)

A2: 31/12/2024 10:38

=day(A2)	31	=hour(A2)	10
=month(A2)	12	=minute(A2)	38
=Year(A2)	2024	=second(A2)	56

Difference blw days.

Start 01/01/24 | End 15/1/24 → =days(End date, start date)

- if end date = today.

A2: 31/12/24 ⇒ days(TODAY(), A2) (today)

Difference blw days considering weekend or holidays

= Networkdays(Start, End, [holidays])

= Networkdays.intl(Start, End, Weekend, [holidays])

Predefined no

Eg:

Networkdays("01/01/2024", "10/10/2024", 8/10/24) ⇒ no. of working days considering holiday at 8/10/24

Networkdays.intl("01/01/24", "10/10/24", 7, 8/10/24) ⇒ no of working day considering weekend + holiday both

7 = predefined weekend no. for Friday & Sat

VLOOKUP

A	B	C	D
FirstName	Price	Vendor	PurchaseMonth
Apple			
Orange			
Kiwi			
Apple			
Pineapple			
Banana			

select all when A in use & 3 = output

G	H	I
Vendor	Fruit Name	Purchase Month
Amazon	Apple	Jan
Blinkit	orange	
Walmart	Pinaple	
BigBasket	Kiwi	

1. =VLOOKUP(A2, G1:H5, 2, FALSE)

Exact value match? → yes → FALSE (3)

2. =VLOOKUP(A3, $G$1:$H$5, 1, False)

(fill)

Vendor | Fruit

For Month: VLOOKUP(A2, $G$1:$I$5, 2, False)

Using True.

Person	Date	Money
	2-7-21	
	19-6-20	
	25-4-18	

Birth	P.Money
1-1-21	400
1-1-20	250
1-1-19	150

= VLOOKUP (B2, $G$1:$H$4, 2, True) → Approx.

& copy to all

⇒ Money
- 400 → 2-7-21 come = or > 1-1-21 (400)
- 250 → 19-6-20 come = or > 1-1-20 (250)
- N/A → 25-4-18 come = or > nothing.

Eg

Numbers	Course
5	N/A
10	MO
25	ME
35	ML

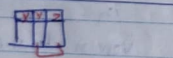
Number	Course
8	MO
16	ME
26	ML

= VLOOKUP (..... True) → & copy all

⇒

Number	Course
1	N/A
10	MO
25	ME
35	ML

Eg:





then go on



Eg: Suppose  in sheet 1

 sheet 2.

= VLOOKUP (A2, Go to sheet 2 → select Range → F4, 2, False/True)

in sheet 2 formula bar enter.



① index = index (select A1 – D4 5, 2, 3) = ans

Row of selected — Column of selected.

② match = match (select X in local table, Range of main table, 0)

cell — (column)

give position of selected entry in main table.

Example:



Nested function.

	A	B	C
	Name	Mark	GRADE
2		85	
3		74	
4		92	
5		69	
6			

Eg

≥ 90	"A"
≥ 80	"B"
≥ 70	"C"
< 70	"L"

=if(C2>=90,"A", else for A
if(C2≥80,"B", else for B
if(C2≥70,"C", else for C
if(C2<70,"L")))

Eg: =sum(A2:A7) (R1)

(R1) ⇒ =sum(B2:B9)

Final R → =sum(A2:A7) & sum(B2:B9)

or.

=product(sum(A2:A7), sum(B2:B9))

CHAT GPT.

text function

→ convert Price column (9.84, 6.87) into Roundoff.

→ Extract Month, Day, AM/PM from Date/Time column

Example: 11/12/24 ⇒ Wednesday / December

16:12 ⇒ PM

→ Make copy of existing column just substitute specific entry Eg: S M L XL | ⇒ | XS M XS XL |

→ Replace existing

Amazon = Replace (A2, 3, 2, K) ⇒ Amkkon

↳ copy formula paste over full column

→ Switch (formula in main table now update / substitute according to extra table / data)

= Switch (A2, J2, K2, A3, J3, K3 - -)

Main table | New value at A2 (in extra table)

Round off

- Roundoff 7.9 → 8 | 8.1 → 8
- Roundup 7.9 → 8 | 8.1 → 9
- Round down 7.9 → 7 | 8.1 → 8.
- Round up to nearest 10 : 492 → 490, 366 → 370

- MROUND (Multiplier)

MROUND without Multip	Multip = 2	= 50.
148.7 → 149	148 (div 2)	150
457.2 → 457	458 (div 2)	450

- MROUND up = ceiling

148.7	150
457.2	500

- MROUND Down = Floor

Statistical formula. ① Large ② Small.

Value column = 10, 8, 17, 9 ⇒

highest value = max func

lowest value = min func.

3rd highest = large (w:w, 3)

3rd lowest = small (w:w, 3)

③ Rank.

A: 10, 8, 17, 9 | B

=Rank (A1, A1:A4)

⇒ 2, 3, 4, 1

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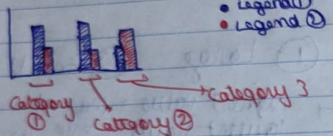
te Graphs:

Best

	Column	
Normal		
Clustered		1 Legend
Stacked		3 Legend

Bar:

	1 Legend
	3 Legend



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Summary:

① Flash fill — Aman Sharma / Divya Chopra → Aman

② Text to column → fixed width / delimiter (/ or space)

③ Sheet Management:
- Auto fit width.
- Freeze unfreeze.
- Copy paste types.
- Merging

④ Filter: Apply new / date F / Time F / Adv. F ① / Adv F ② - Sorting, custom sorting (Multiple sorts in one table)

⑤ Data validation: (Length / date (range) / time (range) / decimal (1.1) / list / Data custom AI Chart GPT - report filter

⑥ Pivot table: Column, Row, Value □, filter, Label filter (Row no), Value filter (Row no), sorting in PT.